YEAR 7 MATHS · CHAPTER 3 · MIND MAP

# Fractions and percentages

tuterly.com.au · Free for students and families

## 1 Introduction to fractions

The bottom says how many equal parts. The top says how many you have.

$\frac{3}{4}$: 3 of 4 equal parts

Improper: $\frac{5}{4}$ is more than a whole

## 2 Equivalent fractions

Multiply or divide the top and bottom by the same number.

$\frac{3}{4} = \frac{6}{8}$

Simplify: $\frac{12}{18} = \frac{2}{3}$

## 3 Mixed and improper

Multiply the whole by the bottom, then add the top.

$2\frac{1}{3} = \frac{7}{3}$

$\frac{11}{4} = 2\frac{3}{4}$

## 4 Ordering fractions

Rewrite with a common denominator, then compare the tops.

$\frac{2}{3}$ and $\frac{3}{4}$: $\frac{8}{12} < \frac{9}{12}$, so $\frac{2}{3} < \frac{3}{4}$

## 5 Adding fractions

Common denominator first, then add the tops only.

$\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$

$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

$1\frac{1}{2} + 2\frac{1}{3} = 3\frac{5}{6}$

**Watch out:** don't add the bottoms: $\frac{1}{2} + \frac{1}{3} \neq \frac{2}{5}$

## 6 Subtracting fractions

Common denominator first, then subtract the tops.

$\frac{3}{4} - \frac{1}{3} = \frac{9}{12} - \frac{4}{12} = \frac{5}{12}$

$2\frac{1}{2} - \frac{3}{4} = \frac{10}{4} - \frac{3}{4} = 1\frac{3}{4}$

## 7 Multiplying fractions

Top times top, bottom times bottom. No common denominator needed.

$\frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5}$

$\frac{1}{2}$ of 12 is 6

## 8 Dividing fractions

Flip the second fraction, then multiply.

$\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times 8 = 6$

$\frac{2}{3} \div 4 = \frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$

## 9 Fractions and percentages

Per cent means out of 100.

$\frac{1}{4} = 25\%$ $\frac{3}{5} = 60\%$

$35\% = \frac{35}{100} = \frac{7}{20}$

## 10 Percentage of a number

Find 10% first, or multiply by the percentage as a fraction.

10% of 60 is 6, so 20% of 60 is **12**

15% of 80: $\frac{15}{100} \times 80 = 12$

## 11 Introduction to ratios

A ratio compares amounts. Order matters.

3 red and 5 blue: red to blue is **3 : 5**

Simplify: $6 : 8 = 3 : 4$

## 12 Problems with ratios

Add the parts, find one part, then multiply.

$40 in the ratio 3 : 5: 8 parts, 1 part = $5

So **$15** and **$25**